

MedAI

One AI Chat. Every Hospital.

An AI-powered mobile platform that collects patient symptoms, recommends the right specialist, and books hospital appointments — all in a single conversation.

Document Type Product Requirements Document (PRD)	Version 1.0.0	Status Final — Hackathon Submission
Platform Flutter (iOS + Android) + React Web Portal	Market Malaysia (Southeast Asia)	

Product Overview

Mission: Remove the friction between a patient's health concern and their first appointment. Replace 20 minutes of phone calls and browser tabs with a 2-minute AI conversation.

2 min Average booking time	100% AI-guided flow	6+ Hospitals on MVP	0 Phone calls needed
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Problem Statement

Healthcare booking in Malaysia is broken. Patients call hospitals during business hours, navigate IVR menus, get transferred, and still may not reach the right department. The result: delayed care, frustrated patients, and missed appointments.

Patient Pain Points

- Don't know which specialist to see
- Manual booking via phone / walk-in
- No single view of all hospitals & availability
- No AI to help triage severity
- Long wait times for confirmation

Hospital Pain Points

- Manual appointment management
- No digital front-door for new patients
- Underutilised specialist schedules
- High no-show rates
- No visibility into patient pipeline

Solution

MedAI is a two-sided platform connecting patients and hospitals through an AI layer.

Patient Mobile App

Flutter app (iOS + Android). AI chat collects symptoms, diagnoses urgency, recommends specialist, and books the appointment — all in one flow.

Hospital Manager Portal

React web portal for hospitals to register, configure their API integration, manage settings, and view appointment analytics.

Core User Flow

1. Open App → 2. Describe Symptoms → 3. AI Collects Anamnesis → 4. AI Verdict → 5. Choose Slot → 6. Confirmed

Target Users

Patient Age 18–65, smartphone users in Malaysia. Wants fast, reliable access to the right doctor without navigating complex hospital systems.	Hospital Manager Administrative staff or IT lead at a private/public hospital. Wants to increase appointment utilisation and reduce manual workload.	Hospital Developer IT engineer integrating the hospital's scheduling system with MedAI via REST API + webhook postbacks.
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Mobile App — Feature Requirements

FR-1: Onboarding

ID	Requirement	Priority
FR-1.1	3-slide onboarding shown on first launch only	P0
FR-1.2	Skip button available on all slides	P1
FR-1.3	State persisted — never shown again after completion	P0

FR-2: Authentication

ID	Requirement	Priority
FR-2.1	Email/password registration with Full Name, Passport, Phone	P0
FR-2.2	Email/password sign-in	P0
FR-2.3	Forgot password email reset	P0
FR-2.4	Session persistence (stay logged in)	P0
FR-2.5	Google Sign-In (optional)	P2

FR-3: AI Chat (Core)

ID	Requirement	Priority
FR-3.1	Hardcoded greeting initiates conversation	P0
FR-3.2	AI collects 6 anamnesis fields via natural conversation	P0
FR-3.3	AI generates specialist recommendation + urgency verdict	P0
FR-3.4	Booking prompt card (Yes / No)	P0
FR-3.5	AI collects booking preferences (budget, hospital, time)	P0
FR-3.6	Inline slot picker cards in chat	P0
FR-3.7	Appointment created in Firestore on slot selection	P0
FR-3.8	Confirmation card with appointment details	P0
FR-3.9	Attach medical documents / images to chat	P1
FR-3.10	Reset/new chat button	P1

FR-4: Appointments

ID	Requirement	Priority
FR-4.1	Upcoming tab: pending + confirmed appointments	P0
FR-4.2	Past tab: cancelled + completed	P0
FR-4.3	Cancel appointment with confirmation dialog	P0
FR-4.4	Real-time Firestore stream (no refresh needed)	P0

Hospital Manager Web Portal

Pages & Features

Route	Purpose	Key Features
/login	Authentication	Email/password sign-in
/register	Hospital onboarding	4-step form: Manager info → Hospital info → Documents → Review
/dashboard	Analytics overview	Appointment counts, line chart, pie chart by specialty, date filters
/settings	Account management	Manager profile, hospital info, documents, danger zone
/integration/api-key	API credentials	View/copy/regenerate API key, creation date, last used
/integration/docs	Developer docs	Full API contract with cURL/JS/Python code examples
/integration/test	API tester	Embedded Postman-like UI to test all 5 endpoints

Firestore Data Model

```
/users/{userId}  fullName, passport, phone, email, avatarUrl, createdAt
/appointments/{appointmentId}  userId, hospitalId, hospitalName, specialistId, specialistName, specialty, externalSlotId, externalAppointmentId, dateTime, status: "pending" | "confirmed" | "cancelled" | "completed", statusSource: "medai" | "hospital_postback", createdAt, updatedAt, createdViaAI: bool
/hospitals/{hospitalId}  name, address, workingHours{}, country, defaultLanguage, integration: { apiBaseUrl, medaiApiKeyMasked, postbackEnabled, lastSyncAt }
/hospitals/{hospitalId}/specialists/{specialistId}  fullName, specialty, price, isActive, updatedAt
/slotCache/{hospitalId_yyyymmdd_specialty}  hospitalId, specialty, date, freeSlots[], fetchedAt, expiresAt
/aiChats/{chatId}  userId, messages[], verdict, recommendedSpecialty, appointmentCreated, createdAt
```

Hospital API Contract

MedAI calls hospital-hosted endpoints using a Bearer token. Hospitals also push updates to MedAI via postback.

Method	Endpoint	Purpose
GET	/specialists	All active specialists with specialty & price
GET	/slots?specialty=&date=	Free slots only, never busy
POST	/appointment/confirm	MedAI books a slot
POST	/appointment/cancel	MedAI cancels a booking
POST	/postback	Hospital pushes status updates to MedAI

Success Metrics (KPIs)

Patient Metrics

Metric	Target (3mo)
Booking completion rate	> 65%
Chat to appointment time	< 3 min
D30 retention	> 40%
App Store rating	>= 4.3 stars

Business Metrics

Metric	Target (3mo)
Hospitals onboarded	10
Monthly active patients	1,000
AI-booked appointments	500/mo
Hospital churn rate	< 5%